

ARPITA MALLIK

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Chittagong, Bangladesh

SUMMARY

I am a recent Computer Science and Engineering graduate. My research interests include Large Language Models, Natural Language Processing, Multilingual and Low-resource NLP, Parameter-Efficient Fine-tuning, and Trustworthy AI, especially in terms of factuality of LLMs, their sycophantic tendencies, deception, and safety concerns. I am especially interested in studying and improving the reliability and robustness of generative AI systems.

EDUCATION

• Chittagong University of Engineering & Technology

BSc. in Computer Science & Engineering

◦ GPA: 3.63/4.00 (most recent 78.25 credits: 3.87/4.00)

March 2022 – July 2026

Chittagong, Bangladesh

RESEARCH EXPERIENCE

• Research Intern, ELITE Research Lab

Adv. Md Kishor Morol - Research Intern, Meta

2026 – Present

◦ Building a counterfactual multimodal deception benchmark for vision-language models.

• Shared Tasks and Workshop Publications

2025 – 2026

- Led and contributed to multiple competitive shared tasks across multimodal misogyny detection and Bengali polarization detection.
- Mentored multiple research projects, contributing to workshop papers accepted at ACL Anthology venues such as DravidianLangTech and LREC.

THESIS

• Undergraduate Thesis

An Emotion-Aware Large Language Model-based QA System for Women's Healthcare Information in Bangla

2026 – Present

- Curated a Bengali and code-mixed (Banglish) women's health QA dataset (2,700 pairs) with binary emotion labels (neutral/concerned) across eight topics including PCOS, menstrual health, pregnancy, menopause, and postpartum depression.
- Built an emotion-aware QA system on a QLoRA (4-bit) fine-tuned Gemma-2-9B model, injecting FiLM-based emotion conditioning at decoder layers via forward hooks, driven by a transformer-based emotion classifier.
- Ran controlled ablations to attribute performance gains to specific interventions rather than architectural novelty.
- Extending the work with a trustworthiness evaluation using 102 health-myth minimal pairs with neutral vs. concerned framing and ground-truth stance labels; measuring directional stance flips (correct→wrong vs. wrong→correct) across baseline, LoRA, and FiLM systems, with automated stance labeling validated against human annotations.

CURRENTLY IN PROGRESS

• Counterfactual Multimodal Deception Benchmark

ELITE Research Lab

2026 – Present

- Designing each scenario against a frozen 2×4 schema (Image_Type: Original/Visual_CF × Prompt_Type: Baseline/Incentive/Oversight/Correction), scored across six deception categories: Fabrication, Omission, Sycophancy, Bluffing, Sandbagging, Obfuscation.
- Building out the dataset across Food Safety, Safety & Security, Wildlife & Nature, and Traffic & Transportation domains.

• Survey on Trustworthy Explainable AI (XAI) in Oncology

2026 – Present

- Co-developing a survey on trustworthy XAI in oncology and clinical decision support, structured around four core dimensions: security (adversarial attacks/defenses on explanations), robustness (perturbation and faithfulness evaluation), fairness (disparities across demographic subgroups), and clinical validation (how unstable explanations affect clinician trust).
- Proposing an original five-dimension evaluation framework: explainability, security, fairness, robustness, and clinical validation.

PUBLICATIONS

C=CONFERENCE, J=JOURNAL, S=IN SUBMISSION, T=THESIS

- [C.1] **CUET-823@DravidianLangTech 2025: Shared Task on Multimodal Misogyny Meme Detection in Tamil Language.** In *Proceedings of the Fifth Workshop on Speech, Vision, and Language Technologies for Dravidian Languages*.
- [C.2] **CUET-823 at SemEval-2026 Task 9: LoRA-Based Instruction-Fine-Tuned LLMs vs. Transformer Models for Bengali Polarization Detection.** Accepted to the *20th International Workshop on Semantic Evaluation (SemEval-2026)*, 2026.

PREPRINTS

- [P.1] **823-OLT@BUET DL Sprint 4.0: Context-Aware Windowing for ASR and Fine-Tuned Speaker Diarization in Bengali Long Form Audio.** *arXiv preprint arXiv:2602.21183*, 2026.
- [P.2] **Assessing the Reliability of Large Language Models in the Bengali Legal Context: A Comparative Evaluation Using LLM-as-Judge and Legal Experts.** *arXiv preprint arXiv:2511.05627*, 2025.

TEACHING EXPERIENCE

- **ASSRO, CUET** 2025 – 2026
Machine Learning Instructor Chittagong, Bangladesh
 - Mentored junior students in Machine Learning and Deep Learning through technical sessions and hands-on workshops.
 - Guided research and implementation for NLP, Computer Vision and Multimodal shared-task projects, contributing to workshop paper publications at ACL Anthology venues.

TECHNICAL SKILLS

- **Programming Languages:** Python, SQL
- **ML/DL Frameworks:** PyTorch, scikit-learn
- **NLP & LLMs:** Transformers, Hugging Face, QLoRA, LoRA, RAG, FiLM Conditioning
- **Deployment & Tools:** FastAPI, Flask, Docker, AWS EC2/S3, Git, GitHub Actions, MLflow, DVC

AWARDS & HONORS

- Finalist – DL Sprint 4.0** (Top 10% / 192 teams, nationwide) February 2026
BUET CSE Fest 2026
- Built a pipeline for Bangla long-form speech recognition and speaker diarization using Whisper-based ASR and fine-tuned diarization models.
- Champion – NodeScape Hackathon** (1st place) July 2025
CUET Computer Club
- Developed an AI-powered graph classification system with algorithm visualization; Dockerized CI/CD deployment using React, Flask, and Python.
- Finalist – Intra CUET Machine Learning Contest** July 2025
IEEE CS CUET Student Branch Chapter
- Built a Bangla physics MCQ question-answering system leveraging LLMs for concept-based queries.
- Top 7 Finalist – HackCSB 2024** (out of 116 teams) August 2024
Coding Shikhe Bangladesh
- Built “MotivationMate,” a mental health-focused web application.

LEADERSHIP & SERVICE

- Vice Chair (Machine Learning Wing)** July 2025 – Present
IEEE CS CUET Student Branch Chapter